

or alternatively as the following minimization problem,

$$\text{Min } L = \frac{E_t(\text{Cost}) - E_0(\text{Cost})}{E_t(\text{TR})}$$

In this case $E_0(\text{Cost})$ is the original cost from the BES in step 2 and $E_t(\text{TR})$ is the price uncertainty for the remainder of the order. If the investor has an alpha expectation this additional cost will need to be incorporated into this expression.

For the trading rate strategy (without an alpha term) the PIM adaptation tactic objective function is:

$$\text{Min } L = \frac{(K_t + (1 - \theta)(b_1 \cdot I' \cdot \alpha_t)) - C^*}{\sigma \cdot \sqrt{(1 - \theta) \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha_t} \cdot P_0}} \quad (8.39)$$

where C^* is the original expected cost, K_t is the unavoidable cost at time t , and the denominator is the price uncertainty for the remainder of the order.

Solving for α_t we get,

$$\alpha_t = \frac{K_t - C^*}{3 \cdot (1 - \theta)(b_1 \cdot I')} \quad (8.40)$$

Incorporation of our boundary conditions yields,

$$\alpha_t^* = \min(\max(LB, \alpha_t), UB)$$

The optimal PIM solution is similar to the optimal AIM solution. The only difference is in the numerator where the PIM numerator is the negative of the AIM numerator. This ensures a mirror image between the two adaptation tactics. As prices become more favorable the PIM tactic will slow down and as prices become less favorable the PIM tactic will speed up. This results in a cost distribution with a higher mean cost than the AIM and targeted cost tactics, but with a much lower probability of incurring higher costs due to persistence of adverse price movement. The PIM tactic protects against the fat tail and “bad” risk events and provides increased opportunity to achieve better prices in times of favorable trends. But this comes with a slightly higher cost. Unfortunately in finance there is no free lunch. This is shown in [Figure 8.3d](#). Notice that the expected cost for PIM, C_4 , is higher than of the constant rate C_1 . But it is associated with less bad risk and a much higher possibility of achieving the better prices from the favorable price trend.